Oxygen Administration

- Few guidelines contain specific instructions on oxygen administration and clinical judgement is required.
- Oxygen should usually only be administered if the patient has an SpO_2 less than 92% on air, airway obstruction, or a specific indication as described within these CPGs.
- Use the simplest device and lowest flow rate required to achieve an SpO_2 of 92-96% (or 88-92% if at risk of hypercarbia) unless specific devices and/or flow rates are described in individual guidelines.
- If pulse oximetry is unreliable or unavailable, administer oxygen as appropriate for the patient's clinical condition.
- The oxygen flow rates usually used are:
  - Nasal prongs 1-4 litres/minute.
  - Simple mask 6 litres/minute.
  - Nebuliser mask 8 litres/minute.
  - Reservoir mask 10 litres/minute.
  - Manual ventilation bag 10 litres/minute.
- If a manual ventilation bag is being used to provide ventilation, use the following PEEP settings:
  - Adult or child:
    - Do not attach PEEP during cardiac arrest.
    - Apply PEEP at 5 cmH_2O for all other conditions. CCPs may choose to increase PEEP if clinically indicated.
    - Refer to the **'cardiogenic pulmonary oedema'** guideline if the patient is an adult and has cardiogenic pulmonary oedema.
  - Neonate:
    - Apply PEEP at 5 cmH_2O, including during cardiac arrest.

Additional information

Oxygen administration

- Oxygen is a treatment for hypoxia and not a general treatment for patients that are ill or injured.
- Oxygen administration is restricted to those patients that have an indication to receive it, because when oxygen levels within blood are higher than normal:
  - Blood vessels (particularly small arteries) vasoconstrict. This has the potential to lower blood flow to tissues and organs.
  - Inflammation is increased and this may worsen inflammatory states.

Oxygen Administration

- Routine oxygen administration during neonatal resuscitation appears to make outcomes worse. This is why oxygen is reserved for deterioration despite initial ventilation.

Delivery devices

- For most patients, nasal prongs or a simple mask is sufficient.
- Reservoir masks should be reserved for patients requiring a high concentration of inspired oxygen. For most patients 10 litres/minute is sufficient oxygen flow through a reservoir mask. The oxygen flow is sufficient if the reservoir bag is not completely deflating.
- Manual ventilation bags should be reserved for patients requiring assistance with their breathing or requiring PEEP. For most patients 10 litres/minute is sufficient oxygen flow through a manual ventilation bag. The oxygen flow is sufficient if the reservoir bag is not completely deflating.
- Ensure a high degree of vigilance if a manual ventilation bag and mask is being held against the face of a small child who is spontaneously breathing, as a small child with respiratory exhaustion may not generate sufficient pressure to open the valves within the manual ventilation bag, and this may result in asphyxiation.

Pulse oximetry

- A pulse oximetry reading does not indicate how well a patient is breathing. How well a patient is breathing is determined by clinical examination and end tidal carbon dioxide ($ETCO_2$) measurement if this is available.
- Pulse oximetry can be unreliable if the patient is very vasoconstricted, shaking, moving, has very dirty fingers, or has been exposed to toxins such as carbon monoxide (CO) or nitrites.
- Do not spend long periods of time trying to obtain a pulse oximetry reading, noting that a common cause of failure to get a reading is severe vasoconstriction.
- Low pulse oximetry readings may be invalid if the SpO_2 waveform on the monitor is damped, flat or irregular. A wide and regular waveform is a sign that the pulse oximetry reading is likely to be valid.

Cyanosis

- A patient may be significantly hypoxic without being cyanosed because cyanosis is usually only detectable with an SpO_2 less than 80%.
- Cyanosis is much more difficult to detect in a patient who is anaemic, has a darker skin tone, or has been exposed to CO.

Oxygen Administration

- Central cyanosis (for example, of the mouth and lips) is usually due to severe hypoxia. Peripheral cyanosis (for example, of the extremities) in the absence of central cyanosis is usually due to vasoconstriction.

Oxygen administration and patients at risk of developing hypercarbia

- Some patients have carbon dioxide clearance that is altered by oxygen administration and excess oxygen administration in these patients may cause hypercarbia.
- Patients at high risk of hypercarbia include patients with COPD, morbid obesity, those on home oxygen and those on home CPAP or BiPAP.
- Oxygen flow rates should be titrated to the patient's known normal SpO_2 if this is known. If this is not known, titrate the oxygen flow rate to an SpO_2 of 88-92%.
- Bronchodilators should be nebulised using air if possible:
  - If using air to nebulise bronchodilators, administer bronchodilators using a flow rate of 8 litres/minute and administer oxygen via nasal prongs if required, adjusting the oxygen flow rate to achieve an SpO_2 of 88-92%.
  - If using oxygen to nebulise bronchodilators, alternating five minutes with the mask on and five minutes with the mask off should only occur if the SpO_2 climbs to above 92% during nebuliser delivery. This is done to limit oxygen exposure whilst delivering most of the nebulised bronchodilator.
- The signs of a rising carbon dioxide level are usually confusion, drowsiness, agitation and a falling level of consciousness. If a patient is suspected of developing hypercarbia, oxygen administration should not be discontinued immediately. Instead, oxygen administration should be reduced to a lower flow rate (targeting an SpO_2 of 88-92%) and the patient reassessed.

Oxygen administration and smoke inhalation

- Oxygen administration is not routinely required following smoke inhalation and should be reserved for patients that have suspected CO poisoning.
- See the **'poisoning from gases'** guideline for more information.

Oxygen administration following bleomycin treatment

- Oxygen administration can cause severe lung inflammation in patients previously treated with bleomycin (a chemotherapy drug) and should usually only be administered to maintain an SpO_2 of 88-92%.
- This sensitivity to oxygen is lifelong following exposure to bleomycin.

Oxygen Administration

Oxygen administration in confined spaces

- Use caution when administering oxygen in a confined space, for example, in a tank, pipe or silo.
- Oxygen administration in a confined space may lead to an increased oxygen concentration within the ambient air. An increase in the oxygen concentration within ambient air to as little as 24% may significantly increase the risk of fire and/or explosion.
- Only administer oxygen within a confined space if:
  - The clinical indication is very strong, and
  - FENZ personnel are present and are monitoring the oxygen concentration within the space.

PEEP (positive end expiratory pressure)

- PEEP improves oxygenation and reduces work of breathing via the following physiological effects:
  - The expiratory pressure reduces lung collapse.
  - The positive pressure reduces venous return to the heart and blood flow through lung vessels, reducing the amount of fluid entering the lungs.
  - The expiratory pressure increases the functional residual capacity. From this more expanded resting position, less work is required for inspiration because of the non-linear compliance of the lungs.
- In cardiac arrest:
  - PEEP is not applied to adults and children during CPR because it reduces the blood flow achieved during CPR. If ROSC is achieved it is appropriate to apply PEEP, but this is not an immediate priority.
  - PEEP is applied to neonates during CPR because the cause of cardiac arrest is usually respiratory failure and the balance of risk is in favour of improving ventilation.
- PEEP reduces cardiac output and should be used with caution in patients showing signs of shock.

More Information

Version history

16 June 2026 1.1.0.1
Update: Added PEEP guidance.

22 November 2022 1.0.1
Update: Indications for oxygen, oxygen administration threshold and guideline simplified.